

Liba A. Snyder

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EDUCATION

State University of New York at Buffalo

Expected Graduation: May 2027

B.S. Mechanical Engineering and B.S. Aerospace Engineering

Buffalo, NY

- Dean's List – GPA: 3.70
- **Relevant Coursework:** Flight Dynamics, Orbital Mechanics, Aerospace Engineering Lab, Elasticity, GD&T, Applied Math

State University of New York at Rockland

Graduated: May 2025

Associate of Engineering Science and Associate of Natural & Physical Sciences

Suffern, NY

- Sam Draper Honors Student, Phi Theta Kappa, President's Society, Biology Club President – GPA: 3.98

EXPERIENCE

WaveShift

January 2026 – Present

Co-Founder

Buffalo, NY

- Co-founded a defense-tech maritime autonomy startup building an 8 m hybrid-electric hydrofoil trimaran USV for ISR.
- Built a parametric Fusion 360 model of the hull, hydrofoils, T-struts, mission bay, and foil-strut interfaces over five design passes.
- Lead product development (mission requirements, payload integration), including a low-cost V0 recon USV prototype and BOM.
- Co-lead pre-seed fundraising: built the investor pitch deck and run investor outreach, grounding market analysis in GAO reports.

The CRASH Lab, University at Buffalo

August 2025 – Present

HERMES Team Lead – Paid Undergraduate Researcher

Buffalo, NY

- Lead a 4-person team developing mechanical systems for HERMES, a Mercury polar ice sample-return mission concept.
- Coordinate subsystem integration across the lander, robotic sample acquisition, cryogenic sample storage, and ascent vehicle.
- Develop CAD concepts, system diagrams, and presentations on subsystem interfaces, design trade-offs, and mission constraints.
- Selected to present HERMES research at AIAA SciTech 2027 (Jan. 2027); co-author of a published AIAA aerospace research paper.

Rockland Industrial Development Agency

April 2025 – August 2025

Research Intern – Grant-Funded Research

Suffern, NY

- Developed a MATLAB GUI integrating circuit models and electrochemical governing equations to optimize electrolyte parameters.
- Conducted cyclic voltammetry to characterize response, sensitivity, and noise, with data analysis in MATLAB and Excel.
- Prepared multi-step electrochemical working electrodes with faculty mentors and presented findings to the Rockland IDA Board.

University at Buffalo

Fall 2026 – Present

Undergraduate Teaching Assistant – MAE 376 Applied Mathematics

Buffalo, NY

- Grade MATLAB-based engineering labs across four lab sections, provide technical feedback, and lead course review sessions.

PROJECTS

Pod Bot – Two-Wheeled Robot & Control App

Personal Project

- Designed a two-wheeled robot in Fusion 360 over four iterations (v1–v4); prepared drawings for patent attorney review.
- Built a multi-robot control app from scratch with VS Code AI agents; users add unlimited robots and control all from one console.

Aluminum Step Ladder – FEA & GD&T

Course Project, MAE 377

- Ran SolidWorks FEA on a 6061-T6 aluminum ladder with a mesh convergence study; evaluated FOS against OSHA Type IA (≥ 3.3).
- Produced GD&T drawings for eight parts per ASME Y14.5-2018, including datum schemes, position at MMC, and profile controls.

TECHNICAL SKILLS

CAD, Design & Manufacturing: SolidWorks, Fusion 360, assemblies, engineering drawings, GD&T (ASME Y14.5), tolerance analysis

Simulation & Analysis: SolidWorks Simulation (FEA), mesh convergence, stress and factor of safety analysis, MATLAB (ODE solvers)

Robotics & Systems: Arduino, robotics prototyping, 3D printing, sensor integration, autonomous systems, mission architecture

Tools & Documentation: MATLAB, Excel, GitHub, LaTeX, AI-assisted development (VS Code coding agents), technical documentation